A New Lower Bound for the Supremum of Autoconvolutions

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Abstract

We establish a lower bound on the autoconvolution constant $C_{1a} = \inf_f \|f * f\|_{L^\infty}/(\int f)^2$, taken over nonnegative $f \in L^1(\mathbb{R})$ supported on $(-1/4, 1/4)$:

$$C_{1a} \;\geq\; 1292/1000 \;=\; 1.292.$$

This improves on the previously announced bound of 1.28 due to Cloninger and Steinerberger [CS17] and on the rigorous analytic bound of 1.27481 established by Matolcsi and Vinuesa [MV10]. The argument extends the Matolcsi–Vinuesa master inequality: the single arcsine kernel is replaced by a convex combination of three arcsine kernels, and the trigonometric multiplier G is re-optimized as a 200-cosine expansion.

The five resulting functionals are evaluated rigorously in `flint.arb` interval arithmetic at 256-bit precision. The no-z_1 quadratic master inequality, applied to the rounded rational anchors, yields strict failure at $M = 1292/1000$ with margin $\tau - \Phi(M) \geq 9.6 \times 10^{-5}$. The analytic reduction is formalized in Lean 4 across 15 modules (≈ 8650 lines of code) rooted at `lean/Sidon/MultiScale.lean`; the headline theorem `autoconvolution_ratio_ge_1292_1000` reaches the three Lean core axioms (`propext`, `Classical.choice`, `Quot.sound`) and exactly two numerical user axioms (`K2_analytic_le_K2UpperQ` and `gain_analytic_ge_gainLowerQ`) that record the certifier's outputs at the rational slacks; the quadratic inversion and the slack-soundness statements are Lean theorems (see Section 5).

1. Introduction

1.1. Overview

This subsection is an informal account of what is proved and why it matters; the precise definitions, the numerical record, and the formal statement of the result follow in Sections 1.2–1.4.

The question. Fix a nonnegative function f on the real line, supported in the short interval $(-1/4, 1/4)$ and normalized so that $\int f = 1$; think of f as a probability density. If X and Y are independent random variables each with density f, then their sum $X + Y$ has density the autoconvolution $f * f$, itself a probability density, now supported in $(-1/2, 1/2)$. Since $f * f$ integrates to 1 over an interval of length 1, its peak value $\|f * f\|_{L^\infty}$ is at least 1, with equality only if $f * f$ were the uniform density on $(-1/2, 1/2)$ — a shape no admissible f produces. The *autoconvolution constant*

$$C_{1a} \;=\; \inf_f \;\|f * f\|_{L^\infty} \Big/ \Big(\int f\Big)^2$$

measures *how flat* an autoconvolution can be made: it is the smallest peak attainable as f ranges over all admissible shapes. Its exact value is unknown; the problem is to trap it between explicit numerical bounds, and in particular to push the lower bound upward.

Why it matters. This continuous extremal problem is the analytic shadow of a classical question in additive combinatorics going back to Erdős and Turán [ET41]: how large can a set of integers be if no integer has too many representations as a sum of two of its elements (a Sidon, or B_2, condition)? Martin and O'Bryant [MO09] made the link precise, showing that the asymptotics of such extremal sets are controlled by this same constant C_{1a}. Each improvement to the *rigorous* lower bound on C_{1a} sharpens the corresponding asymptotic results for $B_2^+[1]$ and related Sidon variants, which is why the constant has drawn sustained attention for several decades.

Why lower bounds are hard. Exhibiting one function f can only ever yield an *upper* bound on the infimum C_{1a}. A *lower* bound is a statement of a different kind: it must certify, simultaneously and for *every* admissible f, that the peak cannot be small. Matolcsi and Vinuesa [MV10] supply exactly such a certificate. From an auxiliary pair chosen by us — a nonnegative "kernel" K whose Fourier coefficients are also nonnegative (a *Bochner-admissible* kernel) together with a trigonometric "multiplier" G — their duality produces a single quadratic inequality that every admissible f must satisfy, and hence a rigorous lower bound on C_{1a}. Any valid pair (K, G) yields *some* bound; the entire game is to construct a pair that yields a *good* one. With a single arcsine kernel and a 119-term cosine multiplier, Matolcsi and Vinuesa obtained $C_{1a} \geq 1.27481$; on the other side, the constant is known to be at most ≈ 1.5032 [GGTW26].

What is new here. We enlarge the class of certificates. Matolcsi and Vinuesa worked with a kernel at a *single* scale. We use that a convex combination of arcsine kernels at *several* scales is still Bochner-admissible (Theorem 3.2), and that mixing scales removes a structural bottleneck. A single-scale arcsine kernel has "blind spots": frequencies at which its Fourier transform nearly vanishes, namely the zeros of a Bessel function. Each near-zero sits in a denominator of the certificate and inflates a critical quantity S_1, throttling the resulting bound. Different scales place their blind spots at different frequencies, so a three-scale blend keeps the Fourier transform uniformly away from zero: S_1 falls from ≈ 87.4 to ≤ 29.841, and the bound rises in proportion. Capturing the gain also requires re-optimizing the multiplier G for the blended kernel, which we do by convex programming with 200 cosine terms. The resulting three-scale construction — arcsine half-widths $(138, 55, 25)/1000$ with weights $(85, 10, 5)/100$ — gives the new lower bound

$$C_{1a} \;\geq\; 1292/1000 \;=\; 1.292,$$

which we call the *PBV bound* (Theorem 1.1).

In what sense this is rigorous. Closing the certificate reduces to controlling five real numbers extracted from (K, G). These are not merely estimated in floating point: each is enclosed in a mathematically guaranteed interval using `flint.arb` arbitrary-precision interval arithmetic at 256-bit precision [Joh17], and each interval is then rounded *outward* to an exact rational. The final quadratic step that produces 1.292 is carried out in exact rational arithmetic, so the conclusion is independent of any floating-point rounding. Independently, the entire analytic reduction — the master inequality, its inversion, and the admissibility of the three-scale kernel — is mechanized in the Lean 4 proof assistant [dMU21, The20] (15 modules, ≈ 8650 lines, no `sorry`). The machine-checked headline theorem rests only on Lean's three logical axioms together with exactly two explicitly identified numerical inputs: decidable inequalities about specific real numbers, the precise analogues of the "Mathematica computed this value" citations in [MV10], but here backed by certified interval arithmetic rather than heuristic numerics. A bundle of standard Fourier-analytic facts is taken as a hypothesis of the general theorem and discharged outright for Schwartz-class f (Section 5.3).

Roadmap. Section 2 recalls the Matolcsi–Vinuesa master inequality and its inversion in the form we need. Section 3 introduces the three-scale arcsine kernel and proves its admissibility. Section 4 certifies the five governing functionals as exact rational bounds via interval arithmetic. Section 5 closes the inequality at the rational witness $M = 1292/1000$, proving the main theorem, and summarizes the Lean formalization.

1.2. The Constant C_{1a}

Let $\mathcal{F} = \{f \in L^1(\mathbb{R}) : f \geq 0,\ \operatorname{supp}(f) \subseteq (-1/4, 1/4),\ \int f > 0\}$. The *autoconvolution constant* is

$$C_{1a} \;=\; \inf_{f \in \mathcal{F}} R(f), \qquad R(f) \;:=\; \frac{\|f * f\|_{L^\infty}}{\left(\int f\right)^2}.$$

By homogeneity we may normalize $\int f = 1$, in which case $R(f) = \|f * f\|_{L^\infty} \geq 1$. Heuristically, $R(f)$ measures the peak height of the self-convolution $f * f$ relative to its L^1 mass, so C_{1a} identifies the flattest autoconvolution achievable in the class $\mathcal{F}$. This constant is the continuous analogue of the Sidon-set extremal problem [ET41, MO09]; improving its lower bound tightens known asymptotic results on $B_2^+[1]$ sets and related Sidon variants.

1.3. Known Bounds

Prior lower bounds on C_{1a} are summarized in Table 1. Matolcsi–Vinuesa [MV10] established the rigorous analytic value 1.27481 in 2010, and Cloninger and Steinerberger [CS17] announced a computational improvement to 1.28 via a discrete branch-and-prune search. All subsequent progress on the constant has been on the upper-bound side, where a recent AI-assisted search lowered the ceiling to 1.5032 [GGTW26]. The present work raises the lower bound to 1.292 (Theorem 1.1), which we refer to as the *Piterbarg–Bajaj–Vincent Bound* (henceforth, the *PBV bound*), by refining the kernel class entering the Matolcsi–Vinuesa dual framework.

Bound	**Value**	**Reference**
Upper bound	≤ 1.5032	Georgiev et al. [GGTW26] (AlphaEvolve)
Upper bound	≤ 1.50972	Matolcsi & Vinuesa [MV10]
Lower bound	≥ 1.262	Martin & O'Bryant [MO09]
Lower bound	≥ 1.27481	Matolcsi & Vinuesa [MV10]
Lower bound	≥ 1.28	Cloninger & Steinerberger [CS17]
Lower bound	≥ 1.292	**This work**

Table 1: Published bounds on C_{1a} in the normalization used here.

1.4. Strategy and Main Result

The Matolcsi–Vinuesa [MV10] dual framework (Section 2) parametrizes a Bochner-admissible kernel K and a cosine multiplier G, producing a quadratic lower bound on $R(f)$ in five real functionals (k_1, K_2, S_1, m_G, a), of which (k_1, K_2) are extracted from the kernel and (S_1, m_G, a) from the multiplier (with $a = (4/u)\, m_G^2/S_1$ a derived gain functional). MV 2010 takes K to be the arcsine kernel at half-width $\delta = 138/1000$ and G a 119-cosine expansion, achieving $R(f) \geq 1.27481$. The present construction (i) replaces the single arcsine by a three-scale convex combination K_{ms}

at half-widths $(138, 55, 25)/1000$ with weights $(85, 10, 5)/100$; (ii) re-optimizes G as a 200-cosine expansion against K_{ms}; (iii) certifies the five functionals in `flint.arb` interval arithmetic; and (iv) solves the no-z_1 quadratic master inequality with the certified rationals. The quadratic inequality $\Phi(M) \geq \tau$ fails strictly at the rational witness $M = 1292/1000$, with margin 9.6×10^{-5} from the rational anchors, giving the main result.

Theorem 1.1 (Main result)**.** *Let $f : \mathbb{R} \to \mathbb{R}$ be nonnegative with* $\mathrm{supp}(f) \subseteq (-1/4, 1/4)$*, $\int f > 0$, and $\|f * f\|_{L^\infty} < \infty$. Then*

$$\frac{\|f * f\|_{L^\infty}}{\left(\int f\right)^2} \;\geq\; \frac{1292}{1000}.$$

In particular $C_{1a} \geq 1292/1000 = 1.292$.

The proof is assembled in Section 5; all numerical anchors are reproducible from the accompanying certificate, and a Lean 4 mechanization is summarized at the end of Section 5.

2. The Matolcsi–Vinuesa Master Inequality

We summarize the dual framework of [MV10, §3]; proofs of the four ingredient lemmas and of the resulting quadratic are recorded in that reference. Throughout, $f \in \mathcal{F}$ with $\int f = 1$, $\widehat{h}(\xi) = \int h(x) e^{-2\pi i \xi x}\, dx$, and $M = R(f) = \|f * f\|_{L^\infty}$. The framework converts the desired lower bound on M into a quadratic inequality whose coefficients depend on two auxiliary objects of our choice, a positive-definite kernel K and a trigonometric multiplier G, so that any rigorous numerical control over (K, G) translates into a rigorous lower bound on M.

2.1. Admissible Kernels and Multipliers

Definition 2.1 (Bochner-admissible kernel)**.** Fix $\delta \in (0, 1/4]$ and set $u = 1/2 + \delta$. A function $K : \mathbb{R} \to \mathbb{R}$ is *Bochner-admissible at scale δ* if

(K1) $K \geq 0$ a.e. on $\mathbb{R}$;

(K2) $K(-x) = K(x)$ for every $x \in \mathbb{R}$;

(K3) $\mathrm{supp}(K) \subseteq [-\delta, \delta]$ and $\int K = 1$;

(K4) the periodic Fourier coefficients $\widetilde{K}(j) := \widehat{K}(j/u)$ satisfy $\widetilde{K}(j) \geq 0$ for every $j \in \mathbb{Z}$.

We write $K_2 := \|K\|^2_{L^2(\mathbb{R})}$.

Definition 2.2 (Admissible cosine multiplier)**.** Fix u and $N \in \mathbb{N}$. A trigonometric polynomial $G(x) = \sum_{j=1}^{N} a_j \cos(2\pi j x/u)$ with $a_j \in \mathbb{R}$ is *admissible* if there is $m_G > 0$ with $G(x) \geq m_G$ for every $x \in [0, 1/4]$. Define

$$S_1 \;=\; \sum_{j=1}^{N} \frac{a_j^2}{\widetilde{K}(j)}, \qquad a \;=\; \frac{4}{u} \cdot \frac{m_G^2}{S_1}. \tag{1}$$

2.2. The Master Inequality and Its Inversion

Theorem 2.3 (Master inequality, z_1-free form)**.** *Let K be Bochner-admissible at scale δ, $u = 1/2 + \delta$, and G admissible of degree N with constant m_G, defining S_1 and a by (1). Then for every $f \in \mathcal{F}$ with $\int f > 0$,*

$$M \;+\; 1 \;+\; \sqrt{(M-1)(K_2 - 1)} \;\geq\; \frac{2}{u} \;+\; a. \tag{2}$$

Proof. Matolcsi–Vinuesa [MV10, Lemma 3.3, Eq. (10)] prove the sharper form

$$M + 1 + 2z_1^2 k_1 + \sqrt{(M - 1 - 2z_1^4)(K_2 - 1 - 2k_1^2)} \;\geq\; \frac{2}{u} + a, \tag{3}$$

where $k_1 = \widehat{K}(1)$ and $z_1 = |\widehat{f}(1)|$ are period-1 Fourier coefficients (so that $\widetilde{K}(j) = \widehat{K}(j/u)$ of Definition 2.1 is the period-u companion). The z_1-*free* designation reflects that z_1 is a Fourier coefficient of the (unknown) extremal f, so it cannot be controlled analytically; the Cauchy–Schwarz step below absorbs z_1 (together with k_1) into the kernel-side quantity K_2, leaving an inequality that depends only on quantities we can certify. Their proof uses the arcsine kernel only through Bochner-admissibility (Definition 2.1); the same derivation therefore applies to any admissible K, in particular to the convex combination K_{ms}. The radicand of (3) is nonnegative by the same Parseval-type bounds as in [MV10, §3]: since f is supported in $[-1/4, 1/4]$, period-1 Plancherel gives $\sum_{j\in\mathbb{Z}} |\widehat{f}(j)|^4 = \|f * f\|_{L^2}^2 \leq \|f * f\|_{L^\infty} = M$, so $2z_1^4 \leq M - 1$ for $f \in \mathcal{F}$ with $\int f = 1$; similarly $\sum_{j\in\mathbb{Z}} |\widehat{K}(j)|^2 = \|K\|_{L^2}^2 = K_2$ gives $2k_1^2 \leq K_2 - 1$ for any Bochner-admissible K with $\int K = 1$. Inequality (2) then follows from (3) by the Cauchy–Schwarz estimate $\sqrt{ac} + \sqrt{bd} \leq \sqrt{(a+b)(c+d)}$ applied to the nonnegative quadruple $(a, b, c, d) = (2z_1^4,\ M - 1 - 2z_1^4,\ 2k_1^2,\ K_2 - 1 - 2k_1^2)$. ☐

Lemma 2.4 (Quadratic inversion)**.** *For fixed $K_2 \geq 1$, the function*

$$\Phi(M) \;:=\; M + 1 + \sqrt{(M-1)(K_2 - 1)}$$

is continuous and strictly increasing on $[1, \infty)$ with $\Phi(1) = 2$. For any $\tau \geq 2$ the equation $\Phi(M) = \tau$ has a unique solution $M_* \in [1, \infty)$, and every $f \in \mathcal{F}$ with $\Phi(R(f)) \geq \tau$ satisfies $R(f) \geq M_*$. In particular, with $\tau = 2/u + a$, $C_{1a} \geq M_*$.

Proof. For $M > 1$, $\Phi'(M) = 1 + \sqrt{(K_2 - 1)/(M - 1)}/2 > 1$, so Φ is strictly increasing on $(1, \infty)$ and continuous at $M = 1$ from the right. Existence, uniqueness, and the conclusion follow from the intermediate value theorem and Theorem 2.3. ☐

The MV 2010 single-arcsine optimum yields $M_* \approx 1.2748$. The remainder of the paper raises M_* by enlarging the kernel class.

3. The Three-Scale Arcsine Kernel

3.1. Single-Scale Arcsine and the Sonine Identity

For $\delta \in (0, 1/4]$, let

$$\eta_\delta(x) \;:=\; \frac{1}{\delta} \cdot \frac{2/\pi}{\sqrt{1 - (2x/\delta)^2}}\, \mathbf{1}_{|x|<\delta/2}$$

denote the arcsine density on $(-\delta/2, \delta/2)$ (the pushforward of the arcsine density on $(-1/2, 1/2)$ under $u \mapsto \delta u$). By the Mehler–Sonine integral representation $J_0(z) = (2/\pi)\int_0^1 \cos(zt)/\sqrt{1-t^2}\,dt$ [DLMF, §10.9.4], $\widehat{\eta_\delta}(\xi) = J_0(\pi\delta\xi)$. The *arcsine kernel at scale* δ is the auto-convolution

$$K_{\mathrm{arc}}(\delta; \cdot) \;:=\; \eta_\delta * \eta_\delta,$$

which is nonnegative and even, supported on $[-\delta, \delta]$, with $\int K_{\mathrm{arc}}(\delta; \cdot) = (\int \eta_\delta)^2 = 1$. The convolution theorem then gives

$$\widehat{K_{\mathrm{arc}}(\delta; \cdot)}(\xi) \;=\; \widehat{\eta_\delta}(\xi)^2 \;=\; J_0(\pi\delta\xi)^2 \;\geq\; 0. \tag{4}$$

In particular each $K_{\mathrm{arc}}(\delta; \cdot)$ is Bochner-admissible at scale δ, with periodic coefficients $\widetilde{K_{\mathrm{arc}}}(\delta; j) = J_0(\pi j \delta/u)^2$ for $u \geq 2\delta$.

3.2. The Convex Combination

Definition 3.1 (Three-scale kernel)**.** Fix

$$(\delta_1, \delta_2, \delta_3) \;=\; \big(\tfrac{138}{1000}, \tfrac{55}{1000}, \tfrac{25}{1000}\big), \qquad (\lambda_1, \lambda_2, \lambda_3) \;=\; \big(\tfrac{85}{100}, \tfrac{10}{100}, \tfrac{5}{100}\big), \tag{5}$$

and set $u = 1/2 + \delta_1 = 638/1000$. The *three-scale arcsine kernel* is

$$K_{\mathrm{ms}}(x) \;:=\; \sum_{i=1}^{3} \lambda_i \, K_{\mathrm{arc}}(\delta_i; x), \qquad x \in \mathbb{R}.$$

i	δ_i	λ_i	u	N
1	138/1000	85/100		
2	55/1000	10/100	638/1000	200
3	25/1000	5/100		

Table 2: Parameters of the three-scale arcsine kernel and the cosine multiplier. The period u and the cosine degree N are shared across the three scales.

3.3. Admissibility

Theorem 3.2 (Admissibility)**.** *The kernel K_{ms} is Bochner-admissible at scale $\delta_1 = 138/1000 < 1/4$, with periodic coefficients*

$$\widetilde{K_{\mathrm{ms}}}(j) \;=\; \sum_{i=1}^{3} \lambda_i \, J_0\Big(\frac{\pi j \delta_i}{u}\Big)^2 \;\ge\; 0, \qquad j \in \mathbb{Z}. \tag{6}$$

Proof. We check conditions (K1)–(K4) of Definition 2.1.

- *(K1) Nonnegativity, (K2) evenness.* Each $\lambda_i K_{\mathrm{arc}}(\delta_i; \cdot) \ge 0$ and is even; nonnegative sums of even nonnegative functions are even and nonnegative.
- *(K3) Support and unit mass.* $\operatorname{supp} K_{\mathrm{arc}}(\delta_i; \cdot) \subseteq [-\delta_i, \delta_i] \subseteq [-\delta_1, \delta_1]$ for $i \ge 1$, so $\operatorname{supp} K_{\mathrm{ms}} \subseteq [-\delta_1, \delta_1]$; and $\int K_{\mathrm{ms}} = \sum_i \lambda_i \int K_{\mathrm{arc}}(\delta_i) = \sum_i \lambda_i = 1$.
- *(K4) Fourier positivity.* Linearity of the Fourier transform and (4) give

  $$\widehat{K_{\mathrm{ms}}}(\xi) \;=\; \sum_i \lambda_i J_0(\pi \delta_i \xi)^2 \;\ge\; 0,$$

  hence (6). □

The motivation is structural: in (1), S_1 blows up at frequencies j where the single-scale $J_0(\pi j \delta / u)^2$ is close to a Bessel zero. Mixing three scales shifts the zeros of each summand, so the sum stays bounded away from zero on $1 \le j \le 200$ (numerically $\min_j \widetilde{K_{\mathrm{ms}}}(j) \ge 2.08 \times 10^{-4}$). This drops S_1 from ≈ 87.4 (single-scale) to ≤ 29.841 (Lemma 4.3), and proportionally raises the gain a.

4. The Rigorous Numerical Certificate

4.1. The Five Certified Anchors

For the master inequality applied to (K_{ms}, G) we need rigorous bounds on five real functionals. We replace each by a rational *anchor*: a floor when the functional enters the master inequality on the side that weakens the bound under understatement (i.e. with a $\geq$ sign), a ceiling on the side that weakens it under overstatement ($\leq$), in both cases obtained by outward rounding of the certifier's arb interval enclosure. Anchoring at exact rationals lets the final inversion (Section 5) be carried out in exact rational arithmetic, independent of the floating-point pipeline that produced the enclosures. The cosine multiplier $G(x) = \sum_{j=1}^{N} a_j \cos(2\pi jx/u)$ with $N = 200$ has rational coefficients $a_j \in \mathbb{Q}$ with denominator 10^8, obtained by rounding the solution of the convex QP that minimizes $\sum_j a_j^2/\widetilde{K_{\mathrm{ms}}}(j)$ subject to $G \geq 1$ on a uniform grid of 5001 points in $[0, 1/4]$ (solved by MOSEK). The five lemmas below are arb-interval computations [Joh17]; all certified outputs are rounded outward to exact rationals. Only four of the five anchors (K_2, S_1, m_G, a) enter the z_1-free form (2) used in Section 5; the kernel-mass moment k_1 is recorded because the certifier internally uses the sharper inequality (3) (which sharpens M_{cert} from 1.292 to ≈ 1.29232 before rounding to the rational headline target).

Lemma 4.1 (Kernel-mass moment)**.** $k_1 \;:=\; \widehat{K_{\mathrm{ms}}}(1) \;=\; \sum_{i=1}^{3} \lambda_i J_0(\pi\delta_i)^2 \;\geq\; 9212/10000.$

Proof. At 256-bit precision, `arb.bessel_j(0)` evaluated at exact rational $\pi\delta_i$ returns balls of radius $< 7 \times 10^{-77}$; the certified sum is $k_1 \geq 0.92124658$, with slack $\geq 4.6 \times 10^{-5}$ above 9212/10000. $\square$

Lemma 4.2 (Kernel-energy moment)**.** $K_2 \;=\; \|K_{\mathrm{ms}}\|_{L^2(\mathbb{R})}^2 \;\leq\; 47897/10000.$

Proof. Split $K_2 = 2\int_0^T \widehat{K_{\mathrm{ms}}}(\xi)^2\,d\xi + 2\int_T^\infty \widehat{K_{\mathrm{ms}}}(\xi)^2\,d\xi$ with $T = 10^5$. The bulk integral is computed by arb adaptive Gauss–Legendre quadrature and certified at $4.78882342\ldots$ with arb radius below 10^{-10}. For the tail, the uniform bound $|J_0(z)|^2 \leq 2/(\pi z)$ [Lan00] holds for $z > 0$; since $\xi \geq T = 10^5$ in the tail forces $z = \pi\delta_i\xi \geq \pi(25/1000)(10^5) \approx 7854$, the bound applies, and applied to both Bessel factors gives $J_0(\pi\delta_i\xi)^2 J_0(\pi\delta_j\xi)^2 \leq 4/(\pi^4\delta_i\delta_j\xi^2)$, whence $\widehat{K_{\mathrm{ms}}}(\xi)^2 = \sum_{i,j}\lambda_i\lambda_j J_0(\pi\delta_i\xi)^2 J_0(\pi\delta_j\xi)^2 \leq 4C^2/(\pi^4\xi^2)$ with $C = \sum_i \lambda_i/\delta_i \approx 9.978$. Integrating $\int_T^\infty \xi^{-2}\,d\xi = 1/T$ and doubling for both half-lines, $2\int_T^\infty \widehat{K_{\mathrm{ms}}}^2 \leq 8C^2/(\pi^4 T) \leq 8.19 \times 10^{-5}$. Summing, $K_2 \in [4.78882342, 4.78890519] \subseteq [0, 47897/10000]$, slack $\geq 7.9 \times 10^{-4}$. $\square$

Lemma 4.3 (Multiplier denominator)**.** $S_1 \;=\; \sum_{j=1}^{200} \frac{a_j^2}{\widetilde{K_{\mathrm{ms}}}(j)} \;\leq\; 29841/1000.$

Proof. Each $\widetilde{K_{\mathrm{ms}}}(j)$ is evaluated in arb from (6) with radius $< 10^{-70}$; dividing exact rational squares a_j^2 yields a certified sum $S_1 \leq 29.840907$, slack $\geq 9.3 \times 10^{-5}$. $\square$

Lemma 4.4 (Pointwise lower bound of G)**.** $m_G \;:=\; \min_{x\in[0,1/4]} G(x) \;\geq\; 998/1000.$

Proof. Partition $[0, 1/4]$ into $n = 32768$ closed cells of half-width $r = 1/(8n) \approx 3.81 \times 10^{-6}$. On each cell centred at c, the second-order Taylor expansion of G yields the rigorous arb enclosure

$$G(\mathrm{cell}) \;\subseteq\; G(c) \;+\; G'(c)\cdot[-r, r] \;+\; G''(\mathrm{cell})\cdot[-r^2/2,\, r^2/2],$$

with G'' evaluated as an arb interval over the entire cell from

$$G''(x) \;=\; -\sum_{j=1}^{200} a_j\,(2\pi j/u)^2 \cos(2\pi jx/u).$$

The minimum of the lower endpoints over all n cells gives the certified bound $m_G \geq 0.99997987$, slack $\geq 1.9 \times 10^{-3}$ above 998/1000. $\square$

Lemma 4.5 (Gain functional)**.** $a \;=\; (4/u)\cdot m_G^2/S_1 \;\geq\; 20925/100000.$

Proof. Substituting the rational anchors of Lemmas 4.3 and 4.4 into the definition of a in (1),

$$a \;\geq\; \frac{4}{638/1000} \cdot \frac{(998/1000)^2}{29841/1000} \;=\; \frac{4000 \cdot 998^2}{638 \cdot 29841 \cdot 10^3} \;=\; \frac{3984016000}{19038558000} \;=\; 0.20926\ldots,$$

hence $a \geq 20925/100000$, slack $\geq 1.0 \times 10^{-5}$. (The underlying arb interval evaluation returns the tighter enclosure $a \geq 0.21009214$ when m_G and S_1 are coupled rather than rounded separately; we use only the looser rational floor in what follows.) $\square$

Functional	Symbol	Direction	Rational bound	Decimal
Kernel-mass moment	k_1	$\geq$	9212/10000	0.9212
Kernel-energy	K_2	$\leq$	47897/10000	4.7897
Multiplier denominator	S_1	$\leq$	29841/1000	29.841
Multiplier minimum	m_G	$\geq$	998/1000	0.998
Gain	a	$\geq$	20925/100000	0.20925

Table 3: Certified rational anchors for the three-scale kernel K_{ms} at parameters (5) with $u = 638/1000$ and the 200-cosine multiplier G.

4.2. Rescaling Invariance

Although the proof of Lemma 4.4 produces $m_G \geq 998/1000$, the master inequality is invariant under $G \mapsto G/m_G$: rescaling multiplies both S_1 and m_G^2 by $1/m_G^2$, leaving a unchanged. The certified rational bound $m_G \geq 998/1000$ is therefore sufficient for the conclusion, and the lemma is not sharpened further.

4.3. Reproducibility

All five anchors are produced by the reproducible script and the associated certificate JSON: each entry is the outward-rounded enclosure returned by `flint.arb` at 256-bit precision; full pipeline inputs (the rational a_j, the bisection schedule, the quadrature tolerance) and the auditor logs are included in the accompanying repository.

5. Proof of the Main Theorem

5.1. The Strict-Failure Witness

We close the master inequality using the anchors of Table 3. The strategy is to exhibit a rational *witness* $M_0 = 1292/1000$ for which $\Phi(M_0) < \tau$ holds strictly; combined with the master inequality $\Phi(R(f)) \geq \tau$, this rules out $R(f) \leq M_0$ by strict monotonicity of Φ (Lemma 2.4), forcing $R(f) > M_0$. No analytic optimization over M is needed: the rational M_0 is fixed in advance and the verification is purely arithmetic. Set $\tau := 2/u + a$ and recall $\Phi(M) := M + 1 + \sqrt{(M-1)(K_2-1)}$ from Lemma 2.4. With the certified rationals from Table 3,

$$\tau \;\geq\; \frac{2}{638/1000} + \frac{20925}{100000} \;=\; 3.134796\ldots + 0.20925 \;=\; 3.344046\ldots$$

Proposition 5.1 (Strict failure at the rational witness)**.** *With* $K_2 \leq 47897/10000$ *and* $\tau = 2/u + a \geq 2000/638 + 20925/10^5$,

$$\Phi(1292/1000) \;\leq\; 3.34395 \;<\; 3.344046\ldots \;\leq\; \tau, \tag{7}$$

with margin $\tau - \Phi(1292/1000) \;\geq\; 9.6 \times 10^{-5}$.

Proof. At $M = 1292/1000$, using the certified upper bound $K_2 \leq 47897/10000$,

$$(M-1)(K_2-1) \;\leq\; \frac{292}{1000}\cdot\frac{37897}{10000} \;=\; \frac{11065924}{10^7} \;=\; 1.1065924,$$

and since Φ is increasing in K_2, this upper bound transfers to $\Phi(M)$. The rational bound $(105195/10^5)^2 = 11065988025/10^{10} > 11065924/10^7$ gives $\sqrt{(M-1)(K_2-1)} \leq 105195/10^5 = 1.05195$, so

$$\Phi(1292/1000) \;\leq\; \frac{1292}{1000} + 1 + \frac{105195}{10^5} \;=\; \frac{66879}{20000} \;=\; 3.34395.$$

Combining with $\tau \geq 4267003/1276000 = 3.344046\ldots$ yields

$$\tau - \Phi(1292/1000) \;\geq\; \frac{4267003}{1276000} - \frac{66879}{20000} \;=\; \frac{307}{3190000} \;\geq\; 9.6\times 10^{-5} \;>\; 0.$$

The conclusion is independently re-certified by an arb cell-search bisection at 256-bit precision, which returns the sharper enclosure $M_{\mathrm{cert}} \geq 1.29232$ when the certifier's full arb interval bounds on all five functionals (rather than the rounded rational anchors of Table 3) are used. □

5.2. Assembly

Theorem 5.2 (Restatement of Theorem 1.1)**.** *Let* $f \in \mathcal{F}$ *with* $\|f*f\|_{L^\infty} < \infty$. *Then* $R(f) \geq 1292/1000$*, and hence* $C_{1a} \geq 1292/1000$.

Proof. By homogeneity assume $\int f = 1$, so $M := R(f) = \|f*f\|_{L^\infty} \geq 1$. We assemble in five steps.

(1) *Admissibility.* By Theorem 3.2, K_{ms} is Bochner-admissible at scale $\delta_1 = 138/1000$, with period $u = 638/1000$.

(2) *Anchors.* By Lemmas 4.1–4.5, the multiplier G and the five rational bounds $k_1 \geq 9212/10000$, $K_2 \leq 47897/10000$, $S_1 \leq 29841/1000$, $m_G \geq 998/1000$, and $a \geq 20925/100000$ hold.

(3) *Master inequality.* Theorem 2.3 applied to (K_{ms}, G) yields $\Phi(M) \geq \tau$. Since Φ is increasing in K_2, replacing the unknown K_2 by the certified upper bound 47897/10000 overestimates Φ; the strict failure $\Phi(M_0) < \tau$ at the witness $M_0 = 1292/1000$ therefore implies $\Phi(R(f)) > \Phi(M_0)$, which (by Lemma 2.4) forces $R(f) > M_0$.

(4) *Strict failure.* By Proposition 5.1, $\Phi(1292/1000) < \tau$.

(5) *Conclusion.* Strict monotonicity of Φ on $[1,\infty)$ (Lemma 2.4) together with (3) and (4) forces $M > 1292/1000$. Taking the infimum over $f \in \mathcal{F}$ gives $C_{1a} \geq 1292/1000 = 1.292$. □

5.3. Lean Formalization

The analytic chain above is mechanized in Lean 4 across 15 modules totalling ≈ 8650 lines of code rooted at `lean/Sidon/MultiScale.lean`, which builds cleanly under Mathlib [The20, dMU21] with zero `sorry` tactics and produces the headline theorem `autoconvolution_ratio_ge_1292_1000`. The module layout factors the proof into: `Sidon.{Defs, Bessel, FourierAux, TorusParseval, MVLemmas, MasterFromLemmas, BundleDefs, BundleEq1, BundleEq2Schwartz, BundleEq3Schwartz, BundleEq4, BilinearParseval, MultiScale, MultiScaleSchwartz, SchwartzAtomicDischarge}`. The headline theorem reaches exactly *two* numerical user axioms in its dependency closure (alongside the three Lean core axioms `propext`, `Classical.choice`, `Quot.sound`):

- `K2_analytic_le_K2UpperQ`: the certified upper bound $K_2(K_{\mathrm{ms}}) \le$ `K2UpperQ` $= 47897/10000$ from Lemma 4.2.
- `gain_analytic_ge_gainLowerQ`: the certified lower bound $a \ge$ `gainLowerQ` $= 20925/100000$ from Lemma 4.5.

Both are analogues of *"Mathematica computed this value"* in MV 2010 [MV10], with the underlying numerical work performed here by the `flint.arb` certifier at 256-bit precision (in `delsarte_dual/grid_bound_alt_kernel/`). The three-scale master inequality at the analytic anchors is itself a Lean *theorem* (`MV_master_inequality_from_MV_lemmas`, composed from the four MV-Lemma-3.1 atomic conclusions formalised in `Sidon.MVLemmas`); slack-monotonicity then lifts it to the rational anchors used in the strict-failure step.

Sound definitions of $f \circ f$ and K_{arc}. The Lean formalisation defines the convolutional autocorrelation as $(f \circ f)(x) := \int_{\mathbb{R}} f(t)\, f(x+t)\, dt$ (matching the MV notation [MV10, §3]; this is `autocorr` in `Sidon.FourierAux`), and the arcsine kernel as the autoconvolution $K_{\mathrm{arc}}(\delta, \cdot) := \eta_\delta * \eta_\delta$ of the half-arcsine density η_δ supported on $(-\delta/2, \delta/2)$. This matches the analytic definitions used throughout Sections 2–3 of the present paper and the Python certifier.

Theorem statements. The headline theorem `autoconvolution_ratio_ge_1292_1000` takes a bundle `ExtremiserPrimitives f` packaging the four MV-Lemma-3.1 atomic primitives for (f, K_{ms}) as hypotheses; the Schwartz specialisation `autoconvolution_ratio_ge_1292_1000_schwartz` discharges the bundle from a single record `SchwartzAtomic f_s` of Schwartz-class atomic Fourier identities, and the slim variant `autoconvolution_ratio_ge_1292_1000_schwartz_residual` reduces the latter to the residual `SchwartzAtomicResidual` hypotheses.

Six further statements are Lean *theorems*: the algebraic inversion `master_inequality_M_lower` (corresponding to Proposition 5.1, proved by case analysis on $M \le 1$ versus $M > 1$ using `Real.sqrt` monotonicity) and the five slack-soundness theorems `K_two_upper_bound`, `k_one_lower_bound`, `S_one_upper_bound`, `min_G_lower_bound`, `gain_lower_bound`, each a one-line `norm_num` verification that the rational slack is on the correct side of the certifier-reported decimal recorded in Lemmas 4.1–4.5.

An explicit finiteness hypothesis $\|f * f\|_{L^\infty} < \infty$ appears in the Lean statement because of the `ENNReal.toReal` encoding; this is harmless for the infimum. The support hypothesis is stated as `Function.support` $f \subseteq (-1/4, 1/4)$, i.e. f vanishes pointwise off $(-1/4, 1/4)$; for any $f \in \mathcal{F}$ in the sense of Section 1.2, its representative obtained by zero-extending off the essential support satisfies the pointwise condition and has the same autoconvolution ratio, so the Lean bound transfers to the L^1-equivalence-class formulation of C_{1a} without loss.

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